11. Build the Coupled-Coil Instrument

The course synthesis, inside the learner boundary

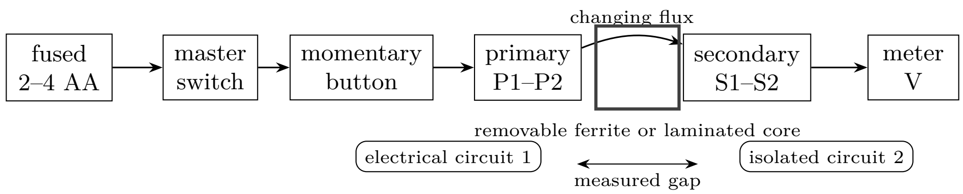
Wind, verify, pulse, and measure a safe magnetic link

Lesson 11 · two careful sessions · learner winds and records; adult inspects and controls power

BOUNDARY: A COUPLED-COIL INSTRUMENT, NOT A TESLA COIL

Use only 2–4 AA cells in a fused, switched holder: 6 V DC maximum. No outlet, mains lead, plug-in transformer, charged capacitor, ignition coil, spark gap, flyback transformer, large battery pack, exposed high voltage, or attempt to make a streamer. The primary receives brief battery pulses. The electrically isolated secondary connects only to a voltmeter or a correctly resistor-limited antiparallel LED pair. Disconnect the holder for every change.

Read the finished instrument



Predict first

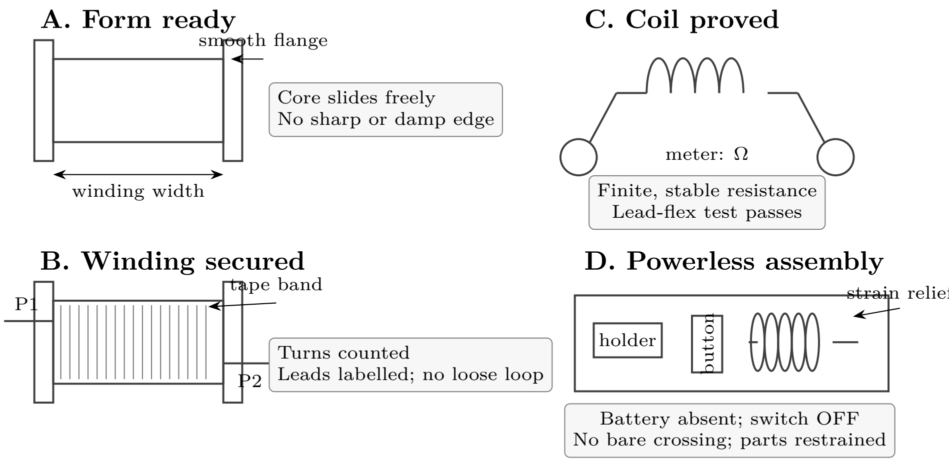
Which change should most increase the receiver indication: aligning the coils, adding the core, or holding steady DC? Mark your order before building. What evidence would make you change your mind?

Gate 0 — freeze the design

Choose two dry cardboard bobbins that accept the same removable core. Plan 100–200 turns per coil and begin with equal counts, so geometry rather than turn ratio is the first variable. Draw and label P1/P2 and S1/S2. Decide where the fuse, OFF label, strain relief, and measurement points will be.

Predict first

Where could the enamel scrape? What prevents a loose loop, a pulled lead, or two bare terminals touching? Point to a control for each failure before touching materials.



Gate 1 — wind, count, and prove each coil

1. With no battery holder on the bench, round or tape every bobbin edge. Measure the winding width and core clearance.
2. Leave a 10 cm lead and tape it as P1. Wind in one direction with neat adjacent turns. Count aloud in tens and put a tally mark on paper for every ten. Do not use a powered drill.
3. Tape the finished layer without hiding either end. Leave 10 cm for P2. Repeat for S1/S2. Record both turn counts.
4. The adult removes enamel from only the last 1 cm of each lead. With the meter on ohms, record R_P and R_S. Gently flex one lead at a time. Each reading must remain finite and stable.

> **Predict first**
>
> If the meter says OL, is the winding continuous? If it reads nearly zero, what accidental short path must be ruled out? Why can a turn count not substitute for this electrical proof?

Gate 2 — assemble with power absent

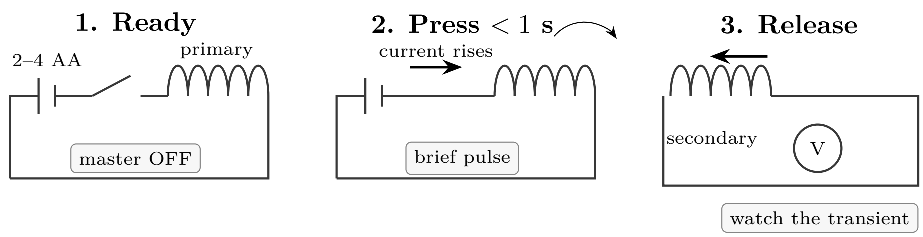
Fasten the holder, master switch, pushbutton, bobbins, and leads to a cardboard base. Add strain relief before joining conductors. Wire holder–fuse–master–button–primary–holder. Connect the secondary only to the voltmeter. Slide the core through both bobbins and confirm that it cannot touch a terminal.

The adult traces both circuits aloud. With cells removed, continuity mode must show: master OFF is open; master ON plus button pressed completes only the primary; P1/P2 has no continuity to S1/S2 in any of the four pairings.

> **Predict first**
>
> Why must the two circuits remain electrically isolated even though energy passes between them? If one P-to-S pairing beeps, list three possible causes.

Gate	Measurement	Pass proof	If it fails
Forms	width; core clearance	smooth; dry; free sliding	remake before winding
Primary	turns; R_P; flex	count and stable finite R_P	clean leads or rewind
Secondary	turns; R_S; flex	count and stable finite R_S	locate break or rewind
Isolation	P-to-S continuity	OL at all four lead pairs	find contact; do not energize
Power path	holder V; OFF/ON state	expected V; OFF is open	adult corrects wiring
Pulse	press/release peak, 3 trials	repeatable indication	follow diagnostic tree

Gate 3 — verify one controlled pulse



Set the secondary meter to a low DC-voltage range. Start with the bobbins coaxial and touching, with the core fully inserted. The adult installs two AA cells, reads the holder voltage, and turns the master ON. The learner presses for less than one second, releases, and records the largest visible indication at both edges. Switch OFF and disconnect the holder. Repeat three times only after confirming holder and coil remain at room temperature.

Predict first

Why can press and release show opposite signs? Why is holding the button down not a stronger induction test? Name every observation that means stop now.

Gate 4 — change one thing

Choose one variable: core in/out, measured gap, coil angle, or cell count within 2–4 AA cells. Hold turn counts, pulse length, meter range, and lead orientation fixed. Make three trials at each setting. Before every reposition: master OFF, holder disconnected, then check for warmth and insulation damage.

Diagnostic tree — no repeatable receiver indication

1. **No source proof:** disconnect coils. Adult checks holder voltage, fuse, master, and button. Do not reconnect until the power path passes.
2. **Primary open:** cells out; remeasure P1–P2 while flexing leads. OL or a jumping value means clean the ends, repair, or rewind.
3. **Receiver open or meter wrong:** remeasure S1–S2; confirm COM/V jacks and volts mode, never the amps jack.
4. **Isolation failed:** check all P-to-S pairs. Any continuity means stop and find the contact or stray strand.
5. **Coupling weak:** return to zero gap, aligned bobbins, core fully inserted, and one brief pulse.
6. **Display missed the transient:** reverse S1/S2 and watch both press and release. A slow meter is a measurement limit, never permission to raise voltage or make sparks.

STOP LINE

Stop for warmth, odor, damaged enamel, a loose lead, a repeatedly opened fuse, or any unexplained result. Do not bypass the fuse. Do not substitute a wall supply, capacitor, large battery, ignition coil, flyback, or spark gap.

Explain from the evidence

A primary current creates a magnetic field. Only a *change* in that current changes the linked flux and induces a secondary voltage, so the useful events are press and release. Alignment, gap, a shared magnetic core, and turn count change how much flux links the windings. This is mutual induction made visible without a shock, spark, or uncontrolled energy store.

Signed acceptance proof

Item	Recorded proof
Primary	turns ________ $R_P =$ ________ Ω flex stable [ ]
Secondary	turns ________ $R_S =$ ________ Ω flex stable [ ]
Isolation	P1–S1 [ ] P1–S2 [ ] P2–S1 [ ] P2–S2 [ ] all OL
Supply / fuse	measured ________ V fuse ________
Baseline geometry	core ________ gap ______ mm
Three press peaks	________________________________
Three release peaks	________________________________
Stop checks	no warmth [ ] no odor [ ] insulation intact [ ]
Learner claim	________________________________
Adult inspection	initials ________ date __________

Extension

Plot peak receiver indication against 0, 5, 10, 20, and 40 mm separation, three trials each. Draw a range bar from smallest to largest trial. Then redraw the accepted wiring and label battery energy, primary current, changing flux, induced voltage, and meter indication.